Professional Online Academy Vu Topper RM

MTH-501 Linear Algebra Update MCQ'S Final Term By Vu Topper RM

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Question No:1 **(Marks:1)** **Vu-Topper RM**

If the augmented matrices of two linear systems are row equivalent, then the two systems have the _____ solution set.

A. same
B. different
C. inconsistant
D. trivial

Question No:2 **(Marks:1)** **Vu-Topper RM**

A m ×n matrix U has orthonormal columns if and only if UT U = I

True
False

Question No:3 **(Marks:1)** **Vu-Topper RM**

Lets={u1+u2+.....up}be the set of non-zero vectors inRnis said to be an orthogonal set if all vectors in S are mutually orthogonal. That is O∉ S andui.uj=0∀ i≠ j, i,j=1,2,……..p.

True
False

Question No:4 **(Marks:1)** **Vu-Topper RM**

If x is orthogonal to both u and v, then x must be _____ to u – v.

A. orthonormal
B. orthogonal
C. equal
D. parallel

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Question No:5 **(Marks:1)** **Vu-Topper RM**

If u=(1, k, 0) and v=(1, 1, 0) are orthogonal vectors then which of the following is the value of k?

A. k = 1

B. k = -1

C. k = 2

D. k = 0

Question No:6 **(Marks:1)** **Vu-Topper RM**

A vector whose length is 1 is called ----------

Unit Vector

Identity Vector

Question No:7 **(Marks:1)** **Vu-Topper RM**

Two vectors are _____if at least one of the vector is a multiple of the other

linearly dependent

linearly independent

Question No:8 **(Marks:1)** **Vu-Topper RM**

If B= is an orthogonal set of vectors with respect to an inner product on a vector space V, then the set B

A. Is an orthonormal basis for v

B. Is linearly independent.

C. Spans the vector space v

D. Linearly dependent.

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Question No:9 **(Marks:1)** **Vu-Topper RM**

If u=(2, 0, -1) and v=(0, -1, 0) is an orthogonal set then which of the following has a norm of 1?

A. v.u
B. u.u
C. v.v
D. u.v

Question No:10 **(Marks:1)** **Vu-Topper RM**

If u and v are vectors in Rn, then we regard u and v as n×1 matrices. The matrix product utv is a ---------- matrix.

A. n × 1
B. n × n
C. 1 × n
D. 1 × 1

Question No:11 **(Marks:1)** **Vu-Topper RM**

7x + 2y = 16
–21x – 6y = 24
The above system has a _____ solution.

A. unique
B. consistent
C. inconsistent
D. trivial

Question No:12 **(Marks:1)** **Vu-Topper RM**

The process of creating the unit vector u from v sometime called--------

Diagonalizing v
Normalizing v

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Question No:13 **(Marks:1)** **Vu-Topper RM**

For u, v vectors in , the distance between u and v is written as
Rn

A. dist(u, v) = ||u − v||
B. dist(u, v) = ||v − v||
C. dist(u, v) = ||u − u||
D. None of these

Question No:14 **(Marks:1)** **Vu-Topper RM**

Suppose x,y,z are some vectors in an inner product space (v,<,>) such that <x,y>=<x,z> for all x∈V, then y=z

True
False

Question No:15 **(Marks:1)** **Vu-Topper RM**

If we divide a non-zero vector by its length we get a

A. zero vector
B. unit vector
C. normalized vector
D. none of these

Question No:16 **(Marks:1)** **Vu-Topper RM**

The matrix A^T(Transpose of A) x A is _____ if and only if the columns of A are linearly independent.

A. symmetric
B. invertible
C. singular
D. scalar

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Question No:17 **(Marks:1)** **Vu-Topper RM**

Let W be a subspace of Rn, then each y in Rn can be written uniquely in the form

A. y = yˆ − z

B. y = yˆ + z

C. y = yˆ/z

D. None of these

Question No:18 **(Marks:1)** **Vu-Topper RM**

Let W be a subspace of , y be any vector in and the orthogonal projection of onto . Then is the closest point in to , in the sense that

Rn Rn yˆ y W yˆ W y

||y − yˆ|| > ||y − v||

||y − yˆ|| < ||y − v||

Question No:19 **(Marks:1)** **Vu-Topper RM**

A square matrix with _____ columns is an orthogonal matrix.

A. same

B. orthogonal

C. orthonormal

D. even

Question No:20 **(Marks:1)** **Vu-Topper RM**

A _____ matrix with orthogonal columns is an orthogonal matrix.

A. rectangular

B. singular

C. square

D. diagonal

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Question No:21 **(Marks:1)** **Vu-Topper RM**

Any orthogonal subsets of vector in an inner product space is linearly independent

True

False

Question No:22 **(Marks:1)** **Vu-Topper RM**

The matrix A^T(Transpose of A) x A is invertible if and only if the columns of A are linearly independent.

True

False

Question No:23 **(Marks:1)** **Vu-Topper RM**

If x is _____ to both u and v, then x must be orthogonal to u – v.

orthonormal

orthogonal

Question No:24 **(Marks:1)** **Vu-Topper RM**

For any vectors u and v, the length of vector u – v will be _____.

A. | u – v |

B. u . v

C. || u – v ||

D. || u . v ||

Question No:25 **(Marks:1)** **Vu-Topper RM**

Any set of three non-coplanar vectors {a,b,c} in ordinary space spans the -------- space

R3

R2

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Question No:26 **(Marks:1)** **Vu-Topper RM**

The matrix A^T(Transpose of A) x A is invertible if and only if the columns of A are _____.

linearly independent

linearly dependent

Question No:27 **(Marks:1)** **Vu-Topper RM**

The two vectors are said to be equivalent if

A. None of the above
B. Both (a) and (b)
C. Same length
D. Same direction

Question No:28 **(Marks:1)** **Vu-Topper RM**

The product of upper triangular matrices is:

A. Diagonal matrix.
B. Upper triangular matrix.
C. Lower triangular matrix
D. None of the above

Question No:29 **(Marks:1)** **Vu-Topper RM**

The _____ vector is orthogonal to every vector in Rn

A. zero vector
B. unit vector
C. normalized vector
D. none of these

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Question No:30 **(Marks:1)** **Vu-Topper RM**
If two rows are orthogonal, they are _____.
linearly independent
linearly dependent

Question No:31 **(Marks:1)** **Vu-Topper RM**
If the columns of a matrix are _____ then the matrix is invertible.
linearly dependent
linearly independent

Question No:32 **(Marks:1)** **Vu-Topper RM**
Let be a five-dimensional vector space, and let be a subset of consisting of five vectors. Then S
Must be linearly independent, but cannot span V
Must be linearly dependent, but may or may not span V

Question No:33 **(Marks:1)** **Vu-Topper RM**
If a _____ matrix has orthonormal columns, then it also has orthonormal rows.
A. singular
B. square
C. rectangular
D. diagonal

Question No:34 **(Marks:1)** **Vu-Topper RM**
If a square matrix has orthonormal columns, then it also has _____ rows.
orthogonal
orthonormal

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Question No:35 **(Marks:1)** **Vu-Topper RM**

A matrix with orthogonal columns is an _____ matrix.

A. identity
B. diagonal
C. orthogonal
D. orthonormal

Question No:36 **(Marks:1)** **Vu-Topper RM**

A square matrix with orthogonal columns is _____ matrix.

A. an orthonormal
B. an invertible
C. an nonivertible
D. an orthogonal

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